

① Dominant kernel: radix-2 Cooley-Turkey butterfly

256-Point FFT has $\log_2(256) = 8$ stages

$$\frac{256}{2} = 128 \text{ butterflies per stage}$$

$$\text{so } 8 \times 128 = 1024 \text{ butterflies total}$$

Data type: INT16 Q1.15

② 1 complex multiply + 2 complex adds per butterfly

$$4 \text{ real multiplies} + 2 \text{ real adds} = 6 \text{ Flops}$$

$$2 \text{ complex adds} : 4 \text{ real adds} = 4 \text{ Flops}$$

$$\text{Total per butterfly} = 10 \text{ Flops}$$

$$= 1024 \times 10 = 10,240 \text{ Flops}$$

③ Lower bound

$$\text{Input } 256 \text{ samples} \times 4 \text{ bytes} = 1024 \text{ bytes}$$

$$\text{Twiddle } 128 \text{ complex} \times 2 \text{ values} \times 4 \text{ bytes} = 1024 \text{ bytes}$$

$$\text{Output } 128 \text{ complex bins} \times 2 \text{ values} \times 4 \text{ bytes} = 1024 \text{ bytes}$$

$$\text{Total} = 3072 \text{ bytes}$$

$$AI = 10,240 / 3072 = 3.33 \text{ Flop / byte}$$

Upper bound:

$$\text{Input} + \text{Output} = 1024 + 1024 = 2048 \text{ bytes}$$

$$AI = 10240 / 2048 = 5 \text{ Flop / byte}$$

④

ASIC

$$\text{ridge} = 1 \text{ GFlops} / 1.6 \text{ GB/s} = 0.625 \text{ Flop / byte}$$

CPU

$$\text{ridge} = 77 / 75 = 1.03 \text{ Flop / byte}$$

Both AI values exceed both ridge points so compute bound

$$\text{ASIC attainable} = 1.0 \text{ GFlops}$$

$$\text{CPU attainable} = 77 \text{ GFlops}$$

